

Supplementary Material

A Physics-Informed Neural Network Forward Solver for the Rough-Volatility Calibration Inverse Problem

This document collects the supporting tables and figures for the manuscript. Every result here is summarised, with its key numbers, in the main text; section references (§) are to the manuscript. All tables are generated from the result files in the public repository by `make_tables.py`, and all figures by `figures_revision.py`.

S1 Compute environment (manuscript §4)

Every run reported in the manuscript and here was performed on the machine below, with no GPU.

Table S1: Compute environment, recorded by `env_report.py`.

Item	Value
Processor	13th Gen Intel Core i5-13450HX
Logical cores	16
Memory (GB)	15.7
Operating system	Windows-11-10.0.26200-SP0
Python	3.12.10
PyTorch	2.12.1+cpu
PyTorch intra-op threads	10
GPU used	no
NumPy	2.5.0
SciPy	1.18.0
BLAS	scipy-openblas 0.3.33.112.0

S2 Liquid-core refit (manuscript §6.2)

The full-surface fit is evaluated unchanged on the restricted points, then the surface is refitted on those points alone from both ends of the H ridge of manuscript §7.5. On both restricted windows the two starts converge to the same parameters, so there H is identified; on the full surface they do not.

S3 Hyperparameter sensitivity (manuscript §7.2)

Two seeds per configuration, 10^4 iterations, scored against a lifted Monte Carlo of 10^6 antithetic paths at 1600 steps. The held-out residual is the RMS PDE residual on a fixed, seeded set of 20000 collocation points never used in training.

Table S2: Liquid-core refit. Fit quality in vol bp, weighted by the vega weights of the objective and unweighted. “Full-surface fit” rows evaluate the reference calibration on the restricted points without refitting; the refits are labelled by the H they started from.

Window	pts	Fit	weighted	unweighted	H	ρ	ν
full	104	full-surface fit	113.2	416.4	–	–	–
$ k \leq 0.10$	53	full-surface fit	160.4	510.8	–	–	–
		refit, start $H=0.17$	146.4	450.0	0.020	–0.360	0.628
		refit, start $H=0.25$	146.6	460.1	0.022	–0.369	0.647
$ k \leq 0.20$	83	full-surface fit	127.0	461.7	–	–	–
		refit, start $H=0.17$	125.0	468.1	0.057	–0.410	0.826
		refit, start $H=0.25$	125.0	467.6	0.057	–0.410	0.826

Table S3: Architecture grid at $n=4$. Solve-error in px bp; held-out PDE residual (RMS); minutes per run on CPU. The minutes were recorded while other jobs shared the machine and are indicative only; the controlled cost is in manuscript Table 3.

width	depth	params	solve-error	residual	min
32	3	2369	23.0 ± 2.1	$3.77e-02$	15.3
32	4	3425	20.9 ± 8.0	$2.73e-02$	20.5
32	5	4481	24.2 ± 0.3	$2.68e-02$	21.6
64	3	8833	20.0 ± 2.6	$2.52e-02$	20.4
64	4	12993	18.2 ± 4.2	$2.08e-02$	25.5
64	5	17153	24.2 ± 2.3	$1.65e-02$	27.5
128	3	34049	11.9 ± 3.2	$1.98e-02$	33.2
128	4	50561	9.7 ± 0.4	$1.36e-02$	33.1
128	5	67073	16.1 ± 2.3	$1.03e-02$	28.2

S4 Signed solve-error by strike (manuscript §7.3)

S5 Profile likelihood (manuscript §7.5)

The values plotted in manuscript Figure 4.

S6 Risk functionals, every vector (manuscript §7.5)

S7 Weekly option-implied parameters (manuscript §7.5)

Table S4: Held-out PDE residual across n (width 64, depth 4, $n_{\text{col}}=2000$), with the solve-error and both quantities relative to $n=4$. The residual is flat to within twelve per cent while the solve-error varies by up to fifty-seven per cent.

n	solve-error	residual	solve/solve ₄	resid./resid. ₄
4	18.2 ± 4.2	$2.08e - 02$	1.00	1.00
8	26.7 ± 7.5	$1.93e - 02$	1.47	0.93
16	28.5 ± 3.1	$2.16e - 02$	1.57	1.04
32	22.1 ± 10.4	$2.00e - 02$	1.22	0.96

Table S5: Collocation ablation at $n=32$ (width 64, depth 4).

n_{col}	solve-error	residual	min
2000	22.1 ± 10.4	$2.00e - 02$	8.7
4000	11.0 ± 7.7	$1.52e - 02$	14.0
8000	11.7 ± 0.6	$1.29e - 02$	35.9

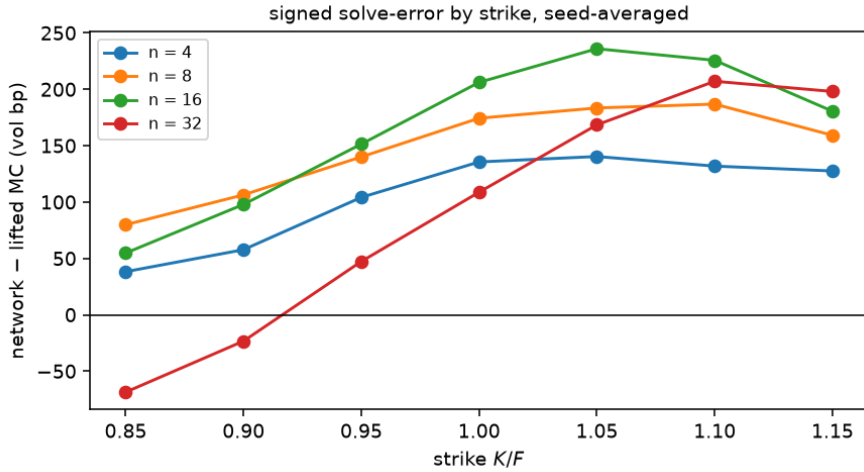


Figure S1: Signed solve-error by strike (network minus lifted Monte Carlo, in implied-vol basis points), seed-averaged over eight seeds, at $n=4, 8, 16, 32$. The network over-prices at and above the money at every n ; the put-wing error is smaller throughout and close to zero at $n=32$.

Table S6: Profile likelihood in H on the headline surface. H is fixed at each grid value and the other five parameters are re-optimised by Nelder–Mead from both ends of the ridge, keeping the better. Weighted and unweighted implied-vol RMSE in vol bp; the last three columns show how ν , ρ and λ compensate along the ridge. From $H=0.30$ upward the re-optimised ν sits on its upper bound of 2.

H	weighted	unweighted	ν	ρ	λ
0.040	114.5	446.7	0.798	−0.421	2.298
0.060	114.1	440.4	0.869	−0.421	2.627
0.090	113.7	432.4	0.984	−0.423	3.155
0.120	113.4	425.5	1.111	−0.424	3.743
0.150	113.3	419.7	1.254	−0.425	4.406
0.180	113.2	415.3	1.414	−0.426	5.154
0.220	113.4	411.9	1.655	−0.428	6.290
0.260	113.8	411.4	1.935	−0.430	7.613
0.300	115.3	430.1	2.000	−0.435	7.720
0.350	120.8	475.9	2.000	−0.444	7.448
0.400	129.6	529.5	2.000	−0.454	7.306
0.450	140.8	582.4	2.000	−0.470	7.378
Best $H = 0.172$; 5% interval $[0.040, 0.300]$.					

Table S7: Risk functionals for every parameter vector the surface cannot tell apart, Fourier-priced at $T=0.25$, $S_0=1$, $r=0$. Variance swap as $\sqrt{K_{\text{var}}}$ in annualised vol points; risk reversal as $\text{IV}(25\Delta \text{ call}) - \text{IV}(25\Delta \text{ put})$ in vol points; collar as long 90% put, short 110% call, in price basis points of spot. Ridge vectors are labelled with their degradation of the weighted objective relative to the best fit.

Vector		Var. swap	25 Δ RR	Collar
reference fit	$H=0.172$	43.47	−6.63	+23.0
profile ridge	$H=0.040$ (+1.1%)	43.13	−6.42	+20.4
profile ridge	$H=0.060$ (+0.8%)	43.17	−6.45	+20.7
profile ridge	$H=0.090$ (+0.4%)	43.25	−6.49	+21.3
profile ridge	$H=0.120$ (+0.2%)	43.32	−6.54	+21.9
profile ridge	$H=0.150$ (+0.0%)	43.40	−6.59	+22.5
profile ridge	$H=0.180$ (+0.0%)	43.49	−6.65	+23.1
profile ridge	$H=0.220$ (+0.2%)	43.61	−6.72	+24.0
profile ridge	$H=0.260$ (+0.5%)	43.73	−6.80	+24.8
profile ridge	$H=0.300$ (+1.9%)	43.69	−6.88	+26.1
hybrid, seed 0	$H=0.071$	43.48	−7.17	+29.9
hybrid, seed 1	$H=0.120$	43.52	−7.19	+30.2
hybrid, seed 2	$H=0.078$	43.42	−7.15	+29.8
range, ridge (9)		0.60	0.46	5.7

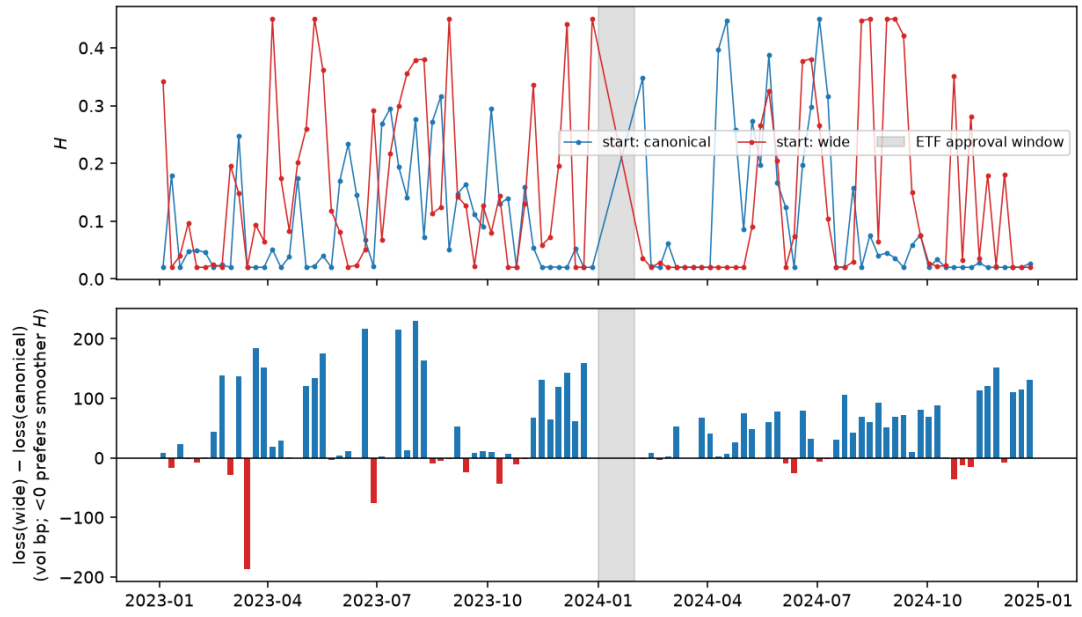


Figure S2: Weekly option-implied H from the trade-tape panel, fitted from both ends of the ridge (top), and which end each week prefers (bottom: loss from the wide-box start minus loss from the canonical start; negative prefers the smoother H). Shaded: January 2024, excluded as the ETF-approval window. The week-to-week scatter and the disagreement between the two starts on the same week are the non-identification of manuscript §7.5 at weekly resolution; the stability test in manuscript Table 5 compares regime means over roughly fifty weeks each.